

1. Define Area Elevation Curve and Capacity Elevation Curves. How does the shape of these curves govern the reservoir simulation in reservoir projects?

Answer:

Area-Elevation Curve:

The Area-Elevation Curve is a graphical relationship between the elevation of the reservoir water surface and the corresponding surface area of the reservoir at that elevation.

Area-Elevation Curve=f (Reservoir Elevation)

It is generally obtained from a topographic survey/contour map of the reservoir basin.

Capacity-Elevation Curve

The Capacity-Elevation Curve shows the relationship between the reservoir water-surface elevation and the volume of water stored in the reservoir at that elevation.

Capacity=f (Reservoir Elevation)

The storage capacity between two elevations can be determined from the corresponding areas using:

$$\Delta S = \frac{A_1 + A_2}{2} (E_2 - E_1)$$

where:

A_1, A_2 = reservoir areas at elevations E_1, E_2

$E_2 - E_1$ = difference in elevation

ΔS = storage volume between the two elevations.

How the Shape of Curves Governs Reservoir Simulation

These curves are very important in reservoir operation and simulation because they establish the relationship between water level, surface area and storage volume.

During simulation, the reservoir water balance is commonly expressed as:

$$S_{t+1} = S_t + I_t - O_t - E_t - L_t$$

where:

S_t = storage at the beginning of the period

I_t = inflow

O_t = release/outflow

E_t = evaporation loss

L_t = other losses.

The curves influence simulation in the following ways:

i. Determination of storage:

Capacity-elevation curve gives the storage corresponding to a particular reservoir level.

- ii. **Determination of water level:**
From the simulated storage, the corresponding reservoir elevation can be obtained.
- iii. **Evaporation losses:**
Area-elevation curve gives the water-surface area required to estimate evaporation loss.
- iv. **Flood routing:**
The curves help determine how much water can be stored and how the reservoir level changes during floods.
- v. **Reservoir operation:**
They help determine whether the reservoir is approaching dead storage, conservation storage or maximum storage.

2. **Enlist the various methods of discharge measurement in rivers. Briefly explain the way of developing stage-discharge and flow duration curves. [5+5=10]**

Answer:

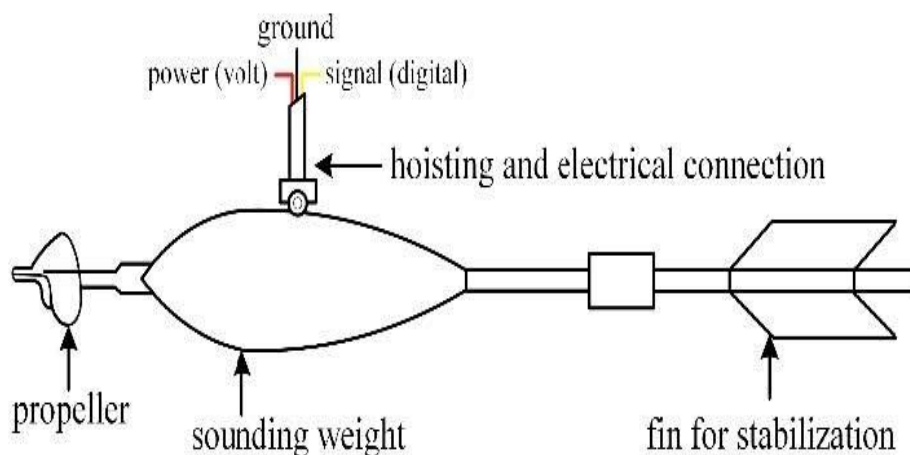
The quantity of water flowing through a stream during a flood, usually expressed in m^3/s is known as flood discharge. Accurate estimation of flood discharge is essential for the design of bridges, culverts, spillways, dams, flood-control structures and other hydraulic works.

The methods for determining discharge in rivers can be broadly classified into direct field measurements and indirect analytical estimations.

i. **Direct Measurement Methods:**

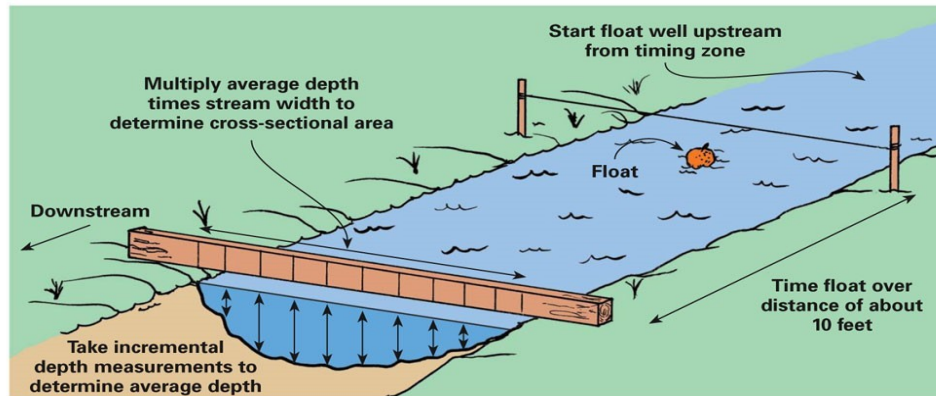
These methods involve measuring physical parameters of the stream flow directly at the site.

- a. **Current Meter (Area-Velocity Method):** This is the standard method for computing discharge. The stream cross-section is divided into several vertical strips, and a current meter is used to measure the velocity (V) at each vertical. The total discharge (Q) is the sum of the discharges from each strip, calculated as: $Q = \sum (\text{Area} \times \text{Velocity})$.



- b. **Float Method:** For simpler or less accurate measurements in small streams, a floating object is timed as it travels a known distance to determine surface velocity.

The Float Method of Estimating Flow



- c. **Weirs and Flumes:** Impervious barriers such as weirs or shaped channels such as flumes are used for small to medium discharges. These structures raise the upstream water level, and the discharge is calculated based on the head of water flowing over the crest.

ii. Indirect and Analytical Estimation Methods

These techniques use historical data, catchment characteristics, or physical evidence of past floods.

- a. **Rational Method:** This is suitable for catchments with limited area. The peak discharge is calculated using the formula: $Q = 360CIA$ Where C is the runoff coefficient, I is rainfall intensity (mm/hr), and A is the catchment area (ha).
- b. **Empirical Formulas:** These are generally used for large catchment areas. Common formulas practiced in Nepal include:
 - a. **Dicken's Formula:** $Q = CA^{3/4}$.
 - b. **Modified Dicken's formula:** $Q = C_t A^{3/4}$
 - c. **Ryve's Formula:** $Q = CA^{2/3}$
- c. **Slope-Area Method:** This indirect technique computes peak flood discharge using high-water marks, channel geometry, and Manning's roughness equation. It is particularly useful for estimating discharges after a flood has passed by measuring the water surface slope and wetted perimeter.
- d. **Unit Hydrograph (UH) Method:** A unit hydrograph represents the direct runoff resulting from a unit depth (1 cm) of excess rainfall occurring uniformly over the basin. It acts as a unit pulse response function to estimate flood hydrographs for various storm durations and intensities.

iii. Continuous Monitoring and Forecasting

- a. **Stage-Discharge (Rating Curve) Method:** A **rating curve** is a graphical relationship between the water level (**stage**) and the corresponding discharge. Once established through multiple field measurements, it allows for continuous discharge monitoring by simply reading the gauge height. The relationship is typically expressed as: $Q = C(H-a)^n$.
- b. **Probable Maximum Flood (PMF):** This is the extreme flood physically possible in a region, determined by maximizing the worst meteorological and hydrological conditions. It is used for designing critical structures where failure would be catastrophic.

iv. Computational Techniques

Modern engineering often employs software for more complex modeling of stream discharge. Frequently used software in Nepal includes HEC-RAS for river modeling, HEC-HMS for watershed analysis, and ArcGIS/QGIS for spatial data assessment.

In hydrology and water-resources engineering, WECS, DHM, and MIP are methods are commonly encountered in Nepal for estimating flood discharge or hydrological design parameters.

i. WECS Method:

WECS stands for Water and Energy Commission Secretariat.

WECS method is an empirical regional method developed for estimating flood discharge in ungauged catchments of Nepal.

It uses catchment characteristics such as catchment area and regional hydrological relationships to estimate design floods.

ii. DHM Method:

DHM stands for Department of Hydrology and Meteorology, Nepal.

The DHM method refers to flood estimation procedures/empirical relationships developed or used by DHM based on observed rainfall and river-flow data from Nepalese gauging stations.

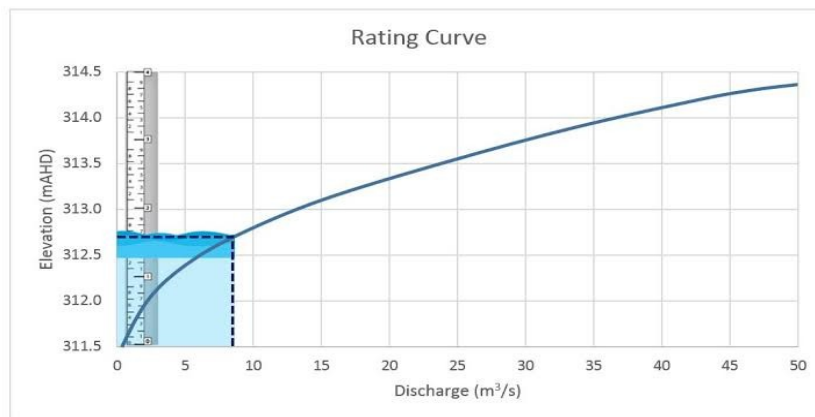
DHM provides hydrological data such as rainfall, river discharge, water levels and flood information, which are used in design.

iii. MIP Method:

MIP generally refers to the Modified Index Flood (MIP) method in regional flood estimation.

The method is based on the index-flood concept, where flood characteristics from gauged catchments are transferred to an ungauged or poorly gauged catchment using regional relationships.

The procedure for developing a stage-discharge curve is described below:



- i. Select a Suitable Gauging Site:
 - a. Straight, stable river reach with a uniform cross-section is selected.
 - b. Bends, backwater effects, and excessive sediment deposition or erosion must be avoided.
- ii. Install a Water Level Gauge
 - a. A staff gauge or automatic water level recorder is installed to measure river stage accurately.
 - b. A permanent reference benchmark should be established.
- iii. Measure Discharge at Different Stages
 - i. Discharge measurements are conducted over a wide range of flow conditions (low, medium, and high flows).
 - b. Velocity-area method with a current meter or an Acoustic Doppler Current Profiler (ADCP) is used.
- iv. Record Corresponding Stage and Discharge
 - a. For each discharge measurement, the corresponding water level (stage) is recorded.
 - b. Sufficient data is collected throughout different seasons.
- v. Plot Stage vs. Discharge
 - a. Stage (H) on the x-axis and discharge (Q) on the y-axis is plotted.
 - b. A smooth best-fit curve is drawn through the plotted points. This curve is called the rating curve.
- vi. Develop the Rating Equation
 - a. Fit an empirical equation to the curve, generally:

$$Q = C(H - a)^n$$

where:

Q = Discharge (m³/s)

H = Gauge height (stage)

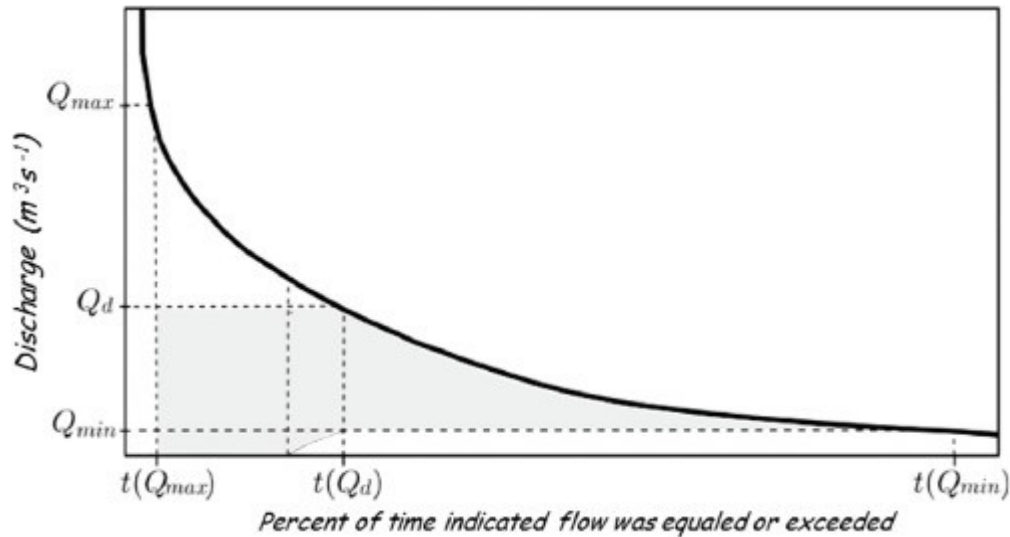
a = Gauge reading corresponding to zero flow

C, n = Empirical constants

vii. Verification and Updating

- a. Periodically the curve is verified by taking new discharge measurements.
- b. The rating curve is revised if changes occur due to scour, sediment deposition, vegetation growth, or changes in the river channel.

A Flow Duration Curve (FDC) is a plot showing the percentage of time that a given discharge is equaled or exceeded. It is widely used to determine the design discharge and hence the installed capacity of a Run-of-River hydropower plant. It is constructed by following procedure:



Procedure:

- i. Collect long-term streamflow data.
- ii. Arrange discharges in descending order.
- iii. Compute exceedance probability.
- iv. Plot the Flow Duration Curve.
- v. Select the design discharge (e.g., Q40, Q50, or Q60).
- vi. Calculate installed capacity using: $P = \rho g Q H \eta$

3. **Water flows in a horizontal channel with velocity of 6 m/s at depth of 1.2 m. Find the conjugate depth and the energy loss on the jump. [10]**

Answer:

Given,

$$V_1 = 6 \text{ m/s}$$

$$y_1 = 1.2 \text{ m}$$

$$g = 9.81 \text{ m/s}^2$$

We Know, Froude's number is given by:

$$Fr_1 = \frac{V_1}{\sqrt{g y_1}}$$

$$Fr_1 = \frac{6}{\sqrt{9.81 \times 1.2}} = 1.75$$

Conjugate depth is given by:

$$\frac{y_2}{y_1} = \frac{1}{2} \left[\sqrt{1 + 8 Fr_1^2} - 1 \right]$$

$$\frac{y_2}{1.2} = \frac{1}{2} \left[\sqrt{1 + 8 (1.75)^2} - 1 \right]$$

$$\frac{y_2}{1.2} = 2.03$$

$$y_2 = 1.2 \times 2.03 = 2.44 \text{ m}$$

Energy loss in hydraulic jump is given by:

$$\Delta E = \frac{(y_2 - y_1)^3}{4 y_1 y_2}$$

$$\Delta E = \frac{(2.44 - 1.2)^3}{4 \times 1.2 \times 2.44}$$

$$\Delta E = 0.162 \text{ m}$$

$$y_2 = 2.44 \text{ m}$$

$$\Delta E = 0.162 \text{ m}$$

4. **What is a run-of-river project? How does it differ from a storage project? Sketch a general arrangement (plan) of such a scheme showing the typical components. [3+3+4=10]**

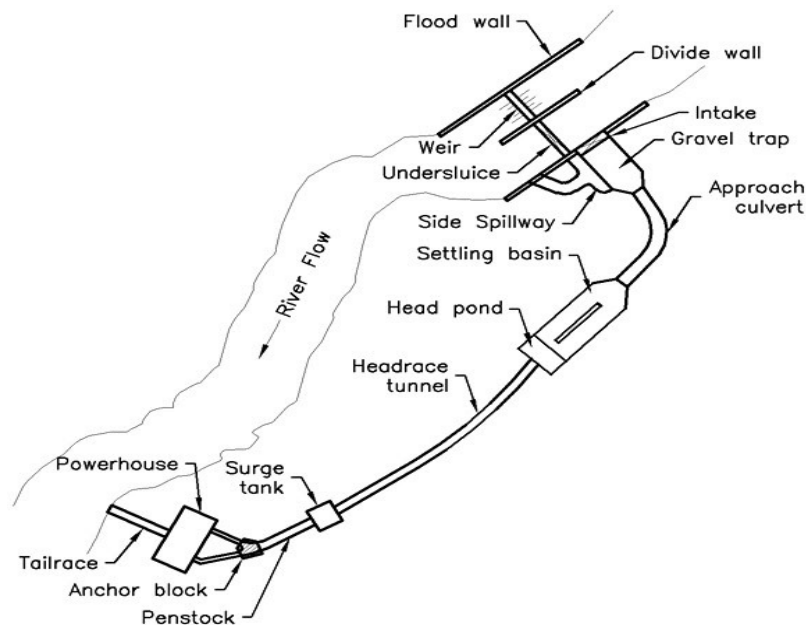
Answer:

RoR hydropower plants do not regulate the hydrograph of the river by one or more seasons. The plant utilizes the minimum flow in a river having no appreciable pondage at its upstream. In this type of plants, weir or barrage construct across a river, simply to rise and maintain the water level at certain height. Usually, low head (up to 70 m) scheme and is suitable only on a perennial river with sufficient dry weather flow to generate designed power. Limited storage is used to meet hourly or weekly function of flow.

Eg: Khimti HPP, Bhotekoshi HPP etc.

It differs from a storage project as given in the points below:

Run-of-River Project	Storage Project
i. Little or no significant storage	i. Large reservoir is provided for storage
ii. Highly dependent on natural river discharge	ii. Less dependent on instantaneous river flow
iii. Varies with seasonal river flow	iii. Can regulate water and generate power according to demand
iv. Generation may decrease considerably	iv. Water stored during wet season can be used in dry season
v. Limited flood-control capacity	v. Can provide significant flood control
vi. Generally smaller	vi. Generally larger due to reservoir
vii. Environmental impact due to ROR plant is usually comparatively lower	vii. Storage project can involve substantial land submergence and displacement.
viii. Generally lower construction cost is required for ROR plants.	viii. Generally higher construction cost due to dam and reservoir for storage plants.
ix. Many Himalayan hydropower projects are designed as RoR schemes	ix. Projects with major reservoirs, such as Budhi Gandaki (proposed), are storage-type



5. **Explain the role of ministry of energy, NEA and DOED for developing of hydropower projects in Nepal and state the importance of Electricity Act, 2049.**

Answer:

Role of the Ministry of Energy, Water Resources and Irrigation:

The Ministry is the **apex government institution** responsible for the overall development and management of Nepal's energy and water-resource sector. Its major roles in hydropower development are:

- i. Formulating national policies, plans, laws, regulations, and standards for hydropower and electricity.
- ii. Preparing short-term and long-term strategies for energy security and hydropower development.
- iii. Establishing an appropriate institutional and investment framework for public, private, foreign, and public-private hydropower projects.
- iv. Providing overall direction for licensing, electricity generation, transmission, distribution, and power trade.
- v. Coordinating with other ministries and agencies concerning forests, land acquisition, environment, finance, industry, local government, and affected communities.

Thus, the Ministry mainly performs the **policy-making, coordination, supervision, and strategic planning functions**.

Role of the Nepal Electricity Authority:

The NEA is the principal public electricity utility of Nepal. Its primary objective is to generate, transmit, and distribute reliable and affordable electricity throughout the country.

Its roles in hydropower development include:

- i. Developing hydropower projects through direct public investment or in cooperation with domestic and foreign investors.
- ii. Operating and maintaining its hydropower stations and associated substations.
- iii. Constructing and managing high-voltage transmission lines and substations.
- iv. Connecting newly developed hydropower projects to the national grid.
- v. Purchasing electricity generated by private hydropower projects through power purchase agreements.

NEA is both a **project developer and the principal off-taker and grid operator**. It is especially important for providing transmission access and creating a market for electricity generated by independent power producers.

Role of the Department of Electricity Development:

The DoED functions under the Ministry and assists it in implementing government policies related to the electricity sector. Its main responsibilities include regulatory administration, licensing, project facilitation, and promotion of private-sector participation.

Its important roles are:

- i. Processing applications for survey, generation, transmission, and distribution licenses.
- ii. Issuing licenses and providing approvals according to the Electricity Act, 2049 and related regulations.
- iii. Providing a “One Window” service for hydropower developers, thereby simplifying administrative procedures.
- iv. Identifying potential hydropower sites and preparing preliminary studies.
- v. Monitoring licensed projects to check compliance with approved designs, schedules, conditions, and legal requirements.
- vi. Supporting transparency and predictability in the electricity regulatory framework.

Therefore, the DoED serves as the regulatory and facilitation arm of the Government. It does not generally function like NEA as a power-generating utility; instead, it enables and regulates project developers.

The Electricity Act, 2049 is the principal legislation governing the survey, generation, transmission, distribution, and supply of electricity in Nepal. Its purpose is to develop electricity by regulating these activities and to standardize and safeguard electricity services. Its importance is as follows:

- i. Provides legal authority for hydropower development:

The Act establishes the legal basis for developing Nepal’s hydropower resources. It clarifies how electricity projects may be surveyed, constructed, operated, and regulated.

- ii. Establishes a licensing system:

A person or corporate body generally cannot conduct the survey, generation, transmission, or distribution of electricity without obtaining the required license. The Act also provides an exemption for certain small projects.

- iii. Encourages private-sector participation:

By allowing individuals and corporate bodies to obtain licenses, the Act opened the electricity sector beyond direct government development. This created a legal foundation for independent power producers, private investment, joint ventures, and foreign participation.

- iv. Regulates the use of water resources for electricity:

Hydropower development involves the use of rivers and other natural resources. The Act gives the Government authority to regulate such use and to impose conditions for the protection of public interests.

- v. Provides a basis for royalties and government revenue:

Hydropower developers are required to pay prescribed fees, royalties, and other charges. This allows the State to receive economic benefits from the use of national water resources and supports revenue sharing and public development.

6. Write short notes on:

a. Plant utilization factor

b. Firm energy and secondary energy

c. General layout of storage power

d. Power Demand Curve

a. Plant utilization factor:

- The Plant Utilization Factor of a hydropower plant is the ratio of the actual energy generated by the plant during a given period to the energy that would have been generated if the plant had operated at its installed capacity continuously during the same period.

- Indicates how effectively the installed generating capacity is utilized.
- A high PUF indicates better utilization of the plant capacity.
- A low PUF may result from seasonal variation in river flow, maintenance, forced outages, or low power demand.

$$\text{Plant Utilization Factor (PUF)} = \frac{\text{Actual Energy Generated}}{\text{Installed Capacity} \times \text{Time Period}}$$

For annual generation:

$$\text{PUF} = \frac{E}{P \times 8760}$$

↳ percentage :

$$\text{PUF}(\%) = \frac{E}{P \times 8760} \times 100$$

Where :

E = Annual energy generated (kWh)

P = Installed plant capacity (kW)

b. Firm energy and Secondary energy:

Firm energy is the amount of electrical energy that a hydropower plant can reliably and continuously generate during the critical low-flow period, when the available water is at its minimum. It is also called dependable or guaranteed energy.

- In a run-of-river project, firm energy is mainly determined by dry-season river flow.
- In a storage project, reservoir storage can be used to maintain firm generation during low-flow periods. Firm energy is generally less than the total annual energy generated by the plant.

Secondary energy is the electrical energy generated by a hydropower plant in excess of its firm energy, usually during periods when sufficient water is available but the energy is not guaranteed throughout the year.

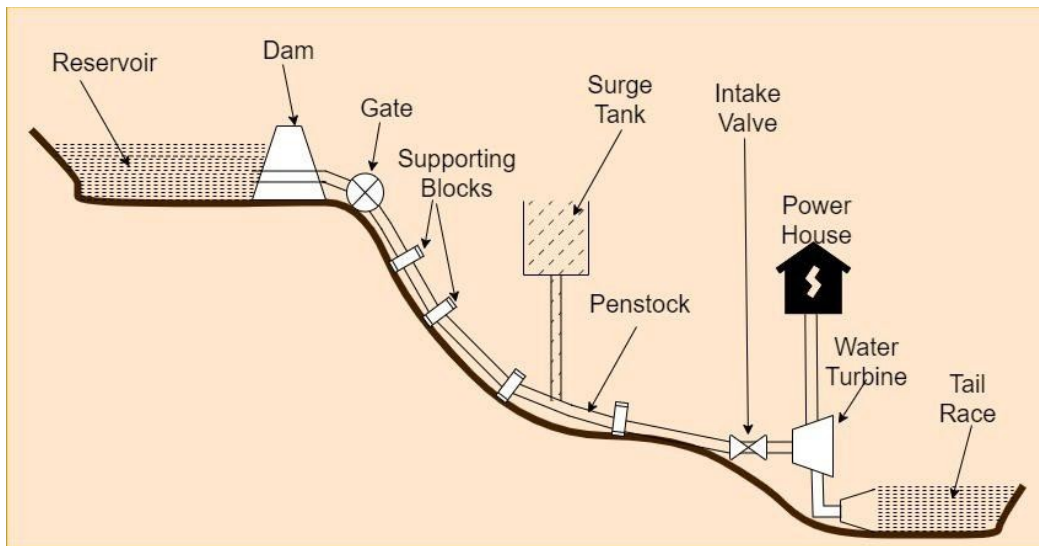
In a run-of-river hydropower plant:

- During the dry season, river flow is low such that the plant produces mainly firm energy.
- During the monsoon, river flow increases and the electricity generation can exceed the firm-energy output. This additional generation is called secondary energy.

Total energy = Firm energy + Secondary energy

c. General layout of storage power:

- Those plants which regulate the hydrograph of the river by one or more seasons are termed as storage plants.
- Powerhouse is located at the foot of dam or away from the dam.
- Water is carried through tunnel to powerhouse.
- It is suitable for medium to high head projects.
- Eg. Kulekhani HPP

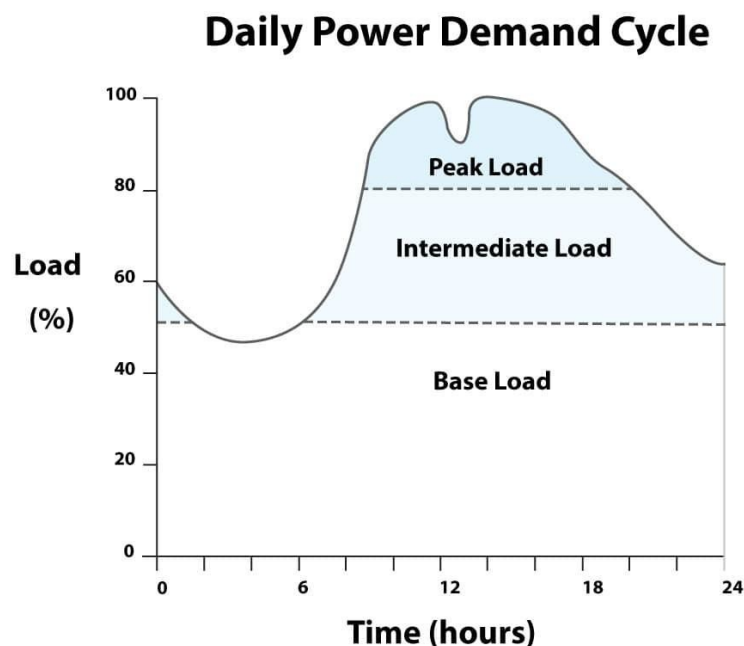


Main Components of storage projects:

- a. **Catchment area:** Area from which rainfall/runoff contributes water to the reservoir.
- b. **Reservoir:** Stores water and regulates its availability throughout the year.
- c. **Dam:** Creates the reservoir and provides the required storage head.
- d. **Spillway:** Safely passes excess flood water downstream.
- e. **Intake structure:** Admits water from the reservoir into the water-conveyance system.
- f. **Headrace tunnel/canal:** Carries water from the intake toward the surge tank or penstock.
- g. **Surge tank:** Controls pressure fluctuations caused by sudden changes in turbine discharge.
- h. **Penstock:** Conveys water under pressure to the turbine.
- i. **Powerhouse:** Contains the turbine, generator and associated equipment.
- j. **Turbine:** Converts hydraulic energy into mechanical energy.
- k. **Generator:** Converts mechanical energy into electrical energy.
- l. **Draft tube:** Carries water from the turbine to the tailrace and helps recover kinetic energy.
- m. **Tailrace:** Returns the water to the downstream river.
- n. **Transformer and switchyard:** Increase the generated voltage and transmit electricity to the grid.

iv. Power demand curve:

A power demand curve is a graphical representation showing the variation of electrical power demand with time over a specified period, such as a day, month, or year.



The curve is plotted with:

- **X-axis:** Time (hours)
- **Y-axis:** Power demand (MW or kW)

The demand varies throughout the day. It is generally low during off-peak hours and increases during periods of high electricity consumption.

7. **What is primary, secondary and exceptional loads on dams? What loads fall into those classification?**

Answer:

The loads acting on a dam are classified into primary, secondary, and exceptional loads according to their importance and frequency of occurrence.

i. Primary Loads:

These are the fundamental loads considered in almost every dam stability analysis. The loads that fall into primary loads are:

- Self-weight (dead load): Weight of the dam structure itself.
- Water pressure: Horizontal hydrostatic pressure exerted by the reservoir water.
- Uplift pressure: Upward water pressure acting beneath the dam foundation.
- Silt/sediment pressure: Lateral pressure exerted by deposited sediment against the upstream face.

ii. Secondary Loads:

These occur depending on environmental and operating conditions. The loads that fall into secondary load classification are:

- Temperature stresses due to variation in temperature.
- Seepage pressure through the dam and foundation.
- Wave pressure due to wind-generated waves.
- Wind pressure on exposed portions of the dam.
- Ice pressure in cold regions.
- Shrinkage and swelling effects of construction materials.

iii. Exceptional Loads:

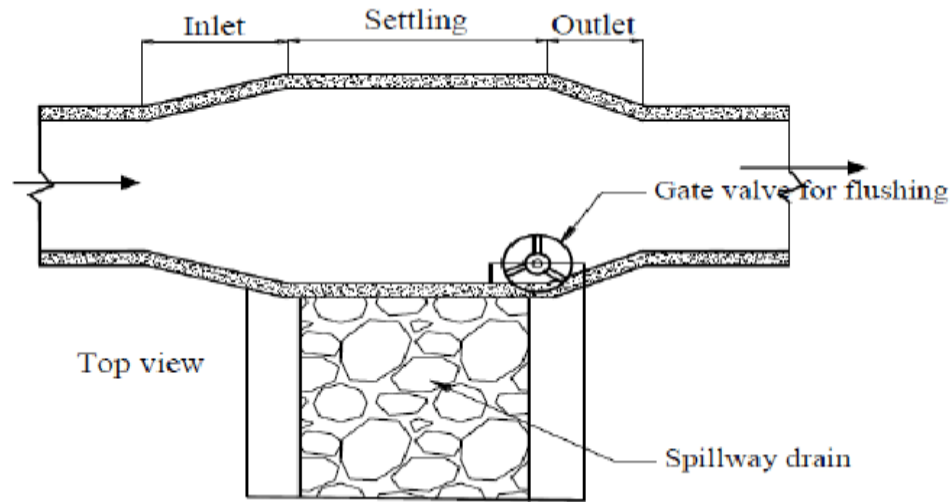
These are rare or extreme loads that may occur during unusual events. The loads that fall under the classification of exceptional loads are:

- Seismic forces
- Exceptional flood loading
- Rapid drawdown of reservoir
- Abnormal sediment loading
- Impact or other accidental loads

8. **Differentiate De-sanding basins with respect to flushing. Mention any one of the methods of designing de-sanding basin.**

Answer:

De-sanding basin also known as de-stilling basin is a hydraulic structure placed between intake and penstock in the hydropower project for removal of suspended sediment particles from flowing water which cannot be trapped by the gravel trap.



De-sanding basins can be classified based on the method used to remove the deposited sediment (flushing) into the following types:

Continuous Flushing De-sanding Basin	Intermittent Flushing De-sanding Basin
i. Sediment is removed continuously during operation.	i. Sediment is removed periodically after accumulation.
ii. A small quantity of water continuously carries deposited sediment through a flushing channel/outlet.	ii. The basin is temporarily taken out of service and the accumulated sediment is flushed using a large quantity of water.
iii. It requires a continuous loss of water for flushing.	iii. It requires a large amount of water only during flushing periods.
iv. It is simple and continuous operation.	iv. It requires periodic operation and control of flushing gates.
v. Little sediment is allowed to accumulate.	v. Considerable sediment accumulates between flushing operations.
vi. Suitable where continuous operation and sufficient water are available.	vi. Suitable where water loss must be minimized and periodic shutdown is acceptable.
vii. Continuous loss of useful water.	vii. Temporary interruption of operation and greater flushing intensity required.

The major design criteria of de-sanding basin are listed below:

i. Design discharge:

- De-sanding basin must be designed for the maximum expected flow rate.
- The capacity of de-sanding basin should be adequate to handle peak flows without reducing its efficiency.

ii. Size of particles:

- The size of particles required to be removed must be selected.
- Generally, particles with its size greater than 0.15-0.2 mm are targeted for removal.

iii. Settling velocity of particles:

- It can be determined using Stoke's law.

$$V_s = \frac{g(\rho_s - \rho)d^2}{18\mu}$$

Where:

V_s = settling velocity

d = particle diameter

ρ_s = density of particle

ρ = density of water

μ = dynamic viscosity

iv. Basin dimensions:

- The length should be adequate for complete settling of particles.
- Length-to-width ratio is generally kept between 4:1 and 8:1.

In a conventional de-sanding basin, the inlet transition is made gradual, whereas the outlet transition can be comparatively steeper.

At the inlet, water enters into de-sanding basin with relatively high velocity. A small inlet angle provides a gradual expansion of the flow section which reduces sudden turbulence, eddies and separation.

At the outlet, the water passed through the settling zone and has relatively low velocity. A larger outlet angle helps to collect the clarifies water and direct it toward outlet.

Hence, inlet angle is provided smaller than outlet angle because the inlet requires more gradual transition to minimize the turbulence and ensure uniform flow distribution.

9. What is surge tank? Mention its functions. Also discuss with reasons the appropriate locations for a surge tank. Present a sketch to clarify your answer. [2+3+5=10]

Answer:

Surge tank is the cylindrical storage tank which is connected to the penstock at suitable point. It is provided in the case of hydropower projects having long penstock pipes. It is located between the headrace pressure conduit and steeply sloping penstock pipe. The main function of surge tank is to protect the long pressure tunnel and penstock pipe from high water hammer effect caused due to sudden acceptance and rejection of loads.

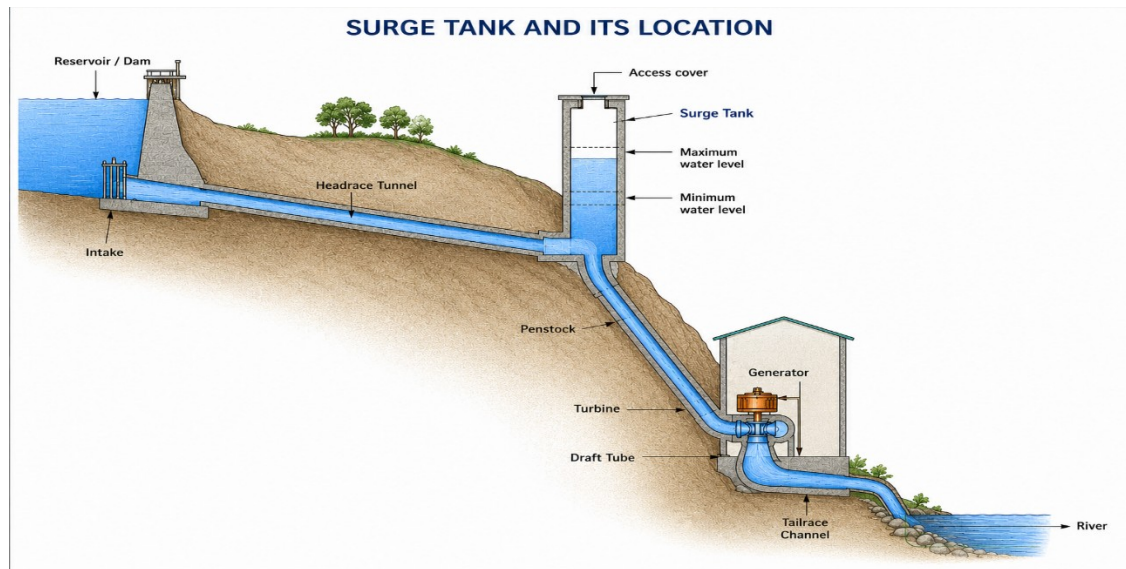
The functions of surge tank are as follows:

- i. **Controls water hammer:**
It reduces excessive pressure rise or fall in the penstock caused by sudden changes in turbine discharge.
- ii. **Supplies additional water:**
When the turbine load increases suddenly, the surge tank supplies additional water until the flow through the headrace tunnel increases.
- iii. **Reduces pressure fluctuations:**
It limits rapid variations in pressure in the water-conductor system.
- iv. **Improves turbine regulation:**
It allows the turbine governor to regulate flow more safely and effectively.
- v. **Protects the penstock:**
By reducing water-hammer effects, it protects the penstock and other hydraulic structures from excessive pressure.

The surge tank is generally located as close as practicable to the powerhouse/turbine and at the junction of the headrace tunnel and penstock.

Reasons for selecting the location:

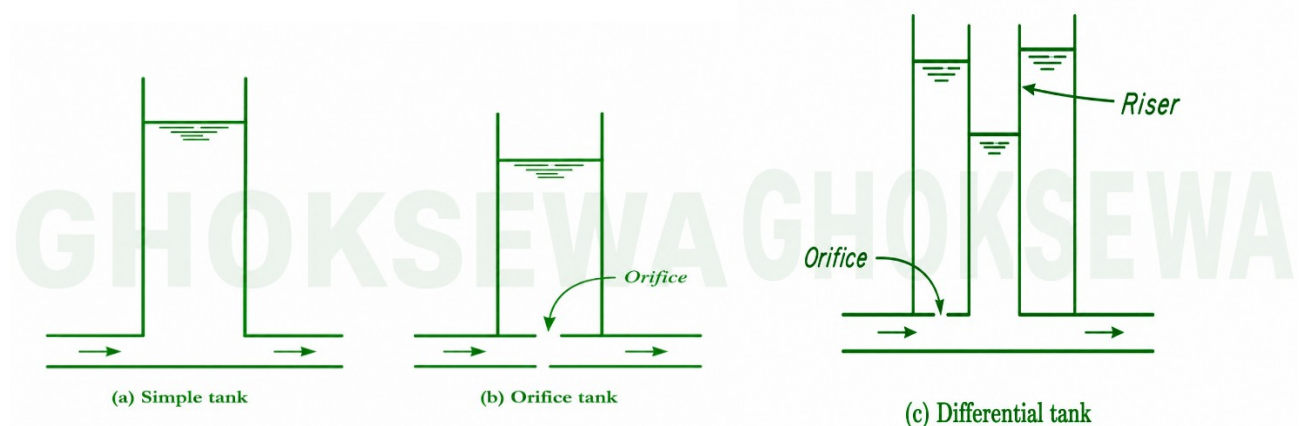
- Shortens the length of penstock exposed to water hammer, thereby reducing pressure fluctuations.
- Provides quick response to sudden changes in turbine discharge.
- The surge tank can readily absorb excess water or supply additional water during load changes.
- It reduces the maximum pressure and pressure-wave travel distance in the penstock.
- It is particularly economical when a long headrace tunnel is followed by a relatively short penstock.



Surge tanks are provided to reduce the water hammer or elastic shock wave coming from the pressure pipeline or tunnel on closing and opening action of the valve.

Surge tank is the cylindrical storage tank which is connected to the penstock at suitable point. It is provided in the case of hydropower projects having long penstock pipes. Surge tank is divided into:

- i. Simple surge tank:
 - It consists of a shaft connected to pressure tunnel directly or by a short connection of cross-sectional area not less than the area of the headrace tunnel.
 - It acts as a reservoir directly connected to the penstock line in such a manner that the water flows into and out of the tank without any head loss.
- ii. Restricted orifice or throttled surge tank:
 - It is a surge tank in which inlet is throttled to improve damping of oscillation by offering greater resistance and connected to the headrace tunnel with or without a connecting shaft.
 - It decreases the amplitude of the surge and effects more rapid reduction if the mass oscillation and gives economy up to 40% in comparison to the simple tank.
- iii. Differential surge tank:
 - It consists of a combination of the simple surge tank and the restricted orifice surge tank except that an internal riser pipe extends upward from the restricted orifice to near the top of the outside shell.
 - It separates out the two functions of simple storing of additional water and effective accelerating and decelerating of water in the main conduit by outside main tank and central river respectively.



10. **Describe layout arrangement of turbines. Discuss important factors that are considered in the selection of turbines. [5+5=10]**

Answer:

Turbine is the hydro-mechanical equipment that converts hydraulic energy into mechanical energy. It consists of wheel called runner provided with the number of curved vanes on the periphery. And guiding apparatus to direct the flow of water at the inlet in specified direction.

Layout arrangements of turbines:

i. **Single turbine arrangement**

- One turbine is connected to one generator.
- Suitable for small hydropower plants or where the discharge is relatively small.
- Simple and economical to operate.

ii. **Multiple turbine arrangement**

- Two or more turbines are installed in parallel.
- Each turbine may operate independently according to the available discharge.
- Provides better efficiency under varying flow conditions and allows maintenance of one unit while others continue operating.

iii. **Horizontal arrangement**

- The turbine shaft is placed horizontally.
- Commonly used for small and medium-head plants, particularly with Francis or Pelton turbines.
- Requires relatively less powerhouse height.

iv. **Vertical arrangement**

- The turbine shaft is vertical, with the generator generally located above the turbine.
- Common in large hydropower stations and low-head plants, especially with Kaplan and large Francis turbines.
- Saves floor space but requires a deeper powerhouse.

The selection of a hydraulic turbine depends on the available head, discharge, power output, efficiency, operating conditions, and economic considerations. Selection of turbine is done such that it gives maximum efficiency, reliable operation, and minimum maintenance cost. The important factors that are considered in the selection of turbines are as follows:

i. **Available head**

- High Head (generally > 250 m) and Low Discharge: Best suited for impulse systems like the Pelton turbine.
- Medium Head (30 m to 250 m) and Medium/High Discharge: Typically utilizes Francis turbines (mixed-flow reaction turbines).
- Low Head (< 30 m) and Large Discharge: Best suited for axial-flow reaction systems like Kaplan or Propeller turbines

ii. **Discharge available**

- The turbine must be capable of handling the available flow efficiently.
- Pelton is suitable for relatively low discharge, whereas Kaplan is suitable for large discharge.

iii. **Specific speed**

- Low Specific Speed (< 50 rpm): Suitable for high-head installations (e.g., Pelton).
- Medium Specific Speed (50 to 260 rpm): Standard for medium-head installations (e.g., Francis).
- High Specific Speed (> 260 rpm): Preferred for low-head, high-discharge systems (e.g., Kaplan/Propeller)

iv. **Efficiency**

- The turbine should have high hydraulic and mechanical efficiency under normal operating conditions.
- The efficiency at part-load conditions is also important where flow varies significantly.

v. **Cavitation**

- The selected turbine should have minimum susceptibility to cavitation.
- Proper turbine setting and draft-tube design are important, particularly for reaction turbines.

vi. **Cost and economy**

- Initial cost, installation cost, operation and maintenance cost, and expected energy generation should be considered.
- The turbine giving the lowest overall life-cycle cost is generally preferred.

In Nepal, Francis turbine is by far the most commonly used turbine type in Nepal's hydropower projects because most of the rivers in Nepal have medium to high heads (roughly 60–350 m) with moderate to high discharge, the conditions where Francis turbines perform best.